Unconstraint

1st NOC $\nabla f(x) = 0$

2nd NOC $\text{Hess} f(x) \geq 0$

2nd SOC $\text{Hess} f(x) > 0$

stationary $\nabla f = 0$

$\text{Hess} f > 0$ minimum

$\text{Hess} f = 0$ minimum

saddle $\leftarrow$ inflection

Constraint

1st NOC KKT Condition

$L = f(x) + vh(x) + ug(x)$

$\nabla_x L = 0$

$u_i g_i(x) \begin{cases} u_i = 0 & g_i(x) \leq 0 \\ g_i(x) = 0 & u_i \geq 0 \quad (\text{active}) \end{cases}$

for regular point

all active $\nabla g_i(x^*)$, $\nabla h(x^*)$

independent

2nd NOC

find general d, all active $\langle \nabla g_i, d \rangle = 0$

$Q = d^T \text{Hess} f(x) d \geq 0$

2nd SOC $Q = d^T \text{Hess} f(x) d > 0$

isolated local minimum

Strong Sufficient Condition : $\text{Hess} f(x) > 0$

KKT x^*

$\text{Hess} L > 0$ isolated minimum

$\langle \nabla g, d \rangle = 0$

$Q = d^T \text{Hess} f(x) d > 0$ isolated minimum

$Q = d^T \text{Hess} f(x) d \geq 0 \begin{cases} \text{minimum} \\ \text{inflection} \end{cases}$

Convex set $x_1, x_2 \in S$, $\alpha x_1 + \beta x_2 \in S$, $\alpha + \beta = 1$

Convex function $f(\alpha x_1 + \beta x_2) \leq \alpha f(x_1) + \beta f(x_2)$, $\alpha + \beta = 1$

$\text{Hess} f(x) \geq 0, \quad \forall x \in S$

Numerical Optimal

$x^{(k+1)} = x^{(k)} + \alpha_k d^{(k)}$

Descent direction

$\langle \nabla f(x^{(k)}), d^{(k)} \rangle < 0$

Stepsize $\alpha = \text{argmin} f(\alpha)$

Convergence : p order

$$\lim_{n \to \infty} \frac{\| x^{(k+1)} - x^* \|}{\| x^{(k)} - x^* \|^p} = c$$

Steepest $d^{(k)} = -\nabla f(x^{(k)})$

Conjugate Gradient $d^{(k)} = -\nabla f(x^{(k)}) + \beta_k d^{(k-1)}$

$\beta_k = [\| \nabla f^{(k)} \| / \| \nabla f^{(k-1)} \|]^2$

Newton $d^{(k)} = -\text{Hess} f(x^{(k)})^{-1} \nabla f(x^{(k)})$

Modified Newton

Solve d : $\text{Hess} f(x^{(k)}) d = -\nabla f(x^{(k)})$ $\alpha = \text{argmin} f(\alpha)$

Quasi - Newton : DFP, BFGS

$d^{(k)} = -A^{(k)} c^{(k)}$

$\alpha_k = \text{argmin} f(\alpha)$

$x^{(k+1)} = x^{(k)} + \alpha_k d^{(k)}$

$A^{(k+1)} = A^{(k)} + B^{(k)} + C^{(k)}$

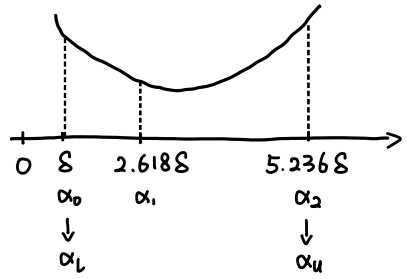
$$B^{(k+1)} = \frac{s^{(k)} s^{(k)T}}{s^{(k)T} y^{(k)}}$$

$$C^{(k)} = \frac{-z^{(k)} z^{(k)T}}{y^{(k)T} z^{(k)}}$$

$s^{(k)} = \alpha_k d^{(k)}$ $y^{(k)} = c^{(k+1)} - c^{(k)}$

$z^{(k)} = A^{(k)} y^{(k)}$

Golden Section : solve unimodel $\min f(\alpha)$ numerically

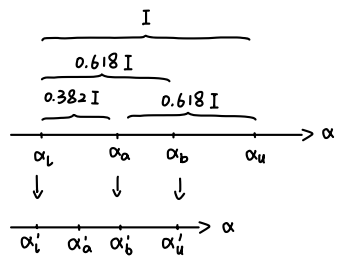


Phase I. α_u. α_l

$$\alpha_q = \delta \sum_{j=1}^{q} (1.618)^q \quad \text{find } q \text{ s.t. } f(\alpha_{q-2}) > f(\alpha_{q-1}) < f(\alpha_q)$$

Phase II. α_a. α_b

$$\tau = 0.618 \begin{cases} \tau^2 = 0.382 \\ 1-\tau = 0.382 \end{cases}$$



if $f(\alpha_a) < f(\alpha_b)$, $\alpha^* \in [\alpha_L . \alpha_b]$

if $f(\alpha_a) > f(\alpha_b)$, $\alpha^* \in [\alpha_a . \alpha_u]$

when $I < \varepsilon$,

$$\alpha^* = \frac{1}{2}(\alpha_L + \alpha_u)$$

Quadratic curve fitting

$\alpha_u . \alpha_l . \alpha_i \rightarrow q(\alpha)$ fitting $f(\alpha)$

$$\bar{\alpha} = \text{argmin } q(\alpha) = -\frac{a_1}{2a_0}$$

$\alpha_i < \bar{\alpha}$	$f(\alpha_i) < f(\bar{\alpha})$	$\alpha^* \in [\alpha_L . \bar{\alpha}]$
	$f(\alpha_i) > f(\bar{\alpha})$	$\alpha^* \in [\alpha_i . \alpha_u]$
$\alpha_i > \bar{\alpha}$	$f(\alpha_i) < f(\bar{\alpha})$	$\alpha^* \in [\bar{\alpha} . \alpha_u]$
	$f(\alpha_i) > f(\bar{\alpha})$	$\alpha^* \in [\alpha_L . \alpha_i]$

GA

Population

Population	$N_p = 4$			Fitness : f value
$x^{(1)}$	1	0	0	16
$x^{(2)}$	0	1	0	4
$x^{(3)}$	0	0	1	1
$x^{(4)}$	1	1	1	49

↑ Chromosome ↑ Gene

(1) Reproduction

fitness $P_i / \Sigma P_i$

$x^{(1)}$	1	0	0	$x^{(1)}$	1	0	0
$x^{(2)}$	0	1	0	$x^{(2)}$	0	1	0
$x^{(3)}$	0	0	1	$x^{(3)}$	0	0	1
$x^{(4)}$	1	1	1	$x^{(4)}$	0	0	1

(2) Crossover

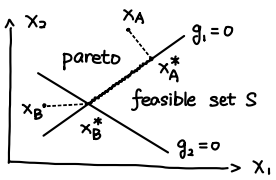
$x^{(2)}$	0	1	0	$x^{(2)}$	0	0	0
$x^{(3)}$	0	0	1	$x^{(3)}$	0	1	1

(3) Mutation

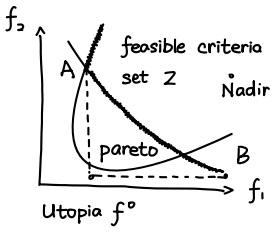
$x^{(1)}$	1	0	0	$x^{(1)}$	1	0	1

Multi - objective

Design Space



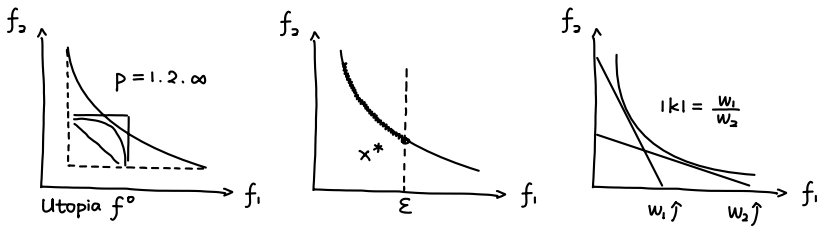
Criteria Space (Cost Space)



Global $\min \Sigma |f_i(x) - f^o|^q$

ε - method $\min f_1(x)$ s.t. $f_2(x) \leq \varepsilon$

Weighted Sum $\min w_1 f_1(x) + w_2 f_2(x)$



Constraint

$\min\ f(x)\quad \text{s.t. } h(x)=0,\ g(x)\le 0$

Penalty $\quad P_h(x) = r h^2(x)$

$$P_g = \begin{cases} r g^2(x), & g \le 0 \\ 0, & g > 0 \end{cases}$$

$P_g = r[g^+(x)]^2 \qquad g^+(x) = \max\{g(x), 0\}$

Barrier $\quad P_g = -\frac{1}{r}\cdot\frac{1}{g(x)} \qquad P_g = -\frac{1}{r}\cdot\log(-g(x))$

ALM $\quad A(x,\lambda,r_p) = f(x) + \lambda_1\Psi(x) + r_p\Psi^2(x) + \lambda_2 h(x) + r_p h^2(x)$

$$x^{(k)} = \arg\min_x A(x,\lambda^{(k-1)}, r_p^{(k-1)}) \qquad \Psi(x) = \max\{g(x), -\frac{\lambda_1}{2r_p}\}$$

$$\lambda_1^{(k)} = \lambda_1^{(k-1)} + 2r_p^{(k-1)} \max\{g(x^{(k)}), -\frac{\lambda_1^{(k-1)}}{2r_p^{(k-1)}}\}$$

$$\lambda_2^{(k)} = \lambda_2^{(k-1)} + 2r_p^{(k-1)} h(x^{(k)})$$

$$r_p^{(k)} = \gamma r_p^{(k-1)}$$

Standard LP

$\min\ f = c^T x$

$\text{s.t. } Ax = b \ge 0,\ x \ge 0$

Transcription

(1) inequality, slack variable (surplus)

(2) unrestricted $\quad x = x^+ - x^-,\ x^+, x^- \ge 0$

Simplex : constraint only "≤" type

$$\begin{array}{ll} \max & x_1 + 2x_2 \\ \text{s.t.} & 2x_1 + x_2 \le 4 \\ & x_1 + 3x_2 \le 6 \\ & x_1, x_2 \ge 0 \end{array} \Rightarrow \begin{array}{ll} \min & -x_1 - 2x_2 \\ \text{s.t.} & 2x_1 + x_2 + x_3 = 4 \\ & x_1 + 3x_2 + x_4 = 6 \\ & x_i \ge 0 \end{array}$$

		-1	-2	0	0		
		x_1	x_2	x_3	x_4	b	Ratio
0	x_3	2	1	1	0	4	4/1 = 4
0	x_4	1	3	0	1	6	6/3 = 2 ✓
	σ	-1	-2 ✓	0	0		

选最负，x_2 入基

选最小正数

x_4 出基

行标准型

3化为1

		-1	-2	0	0		
		x_1	x_2	x_3	x_4	b	Ratio
0	x_3	5/3	0	1	-1/3	2	$2/\frac{5}{3} = 6/5$ ✓
-2	x_2	1/3	$\boxed{1}$	0	1/3	2	$2/\frac{1}{3} = 6$
	σ	-1/3 ✓	0	0	2/3		

$-1 - 0\times\frac{5}{3} - (-2)\times\frac{1}{3}$

		-1	-2	0	0	
		x_1	x_2	x_3	x_4	b
-1	x_1	1	0	3/5	-1/5	6/5
-2	x_2	0	1	-1/5	2/5	8/5
	σ	0	0	1/5	1/5	

全正，唯一最优

1) Ratio 全非正，unbounded

2) σ 非基列出现 3 0，multiple solutions

只有 ≤ 型必 feasible

$x^* = (\frac{6}{5}, \frac{8}{5}, 0, 0) \qquad f^* = -1\times\frac{6}{5} - 2\times\frac{8}{5} = -\frac{22}{5}$

Artifical Variable : constraint "=" , "≥" type
tabeaular no diagnal 1

$$\begin{aligned} \max\ & x_1 + 4x_2 \\ \text{s.t.}\ & x_1 + 2x_2 \le 5 \\ & 2x_1 + x_2 = 4 \\ & x_1 - x_2 \ge 1 \end{aligned} \Rightarrow \begin{aligned} \min\ & -x_1 - 4x_2 \\ \text{s.t.}\ & x_1 + 2x_2 + x_3 = 5 \\ & 2x_1 + x_2 = 4 \\ & x_1 - x_2 - x_4 = 1 \end{aligned} \xRightarrow{\text{AV}} \begin{aligned} \min\ & -x_1 - 4x_2 \\ \text{s.t.}\ & x_1 + 2x_2 + x_3 = 5 \\ & 2x_1 + x_2 + x_5 = 4 \\ & x_1 - x_2 - x_4 + x_6 = 1 \end{aligned}$$

Phase I : min $w = x_5 + x_6$

$w^* = 0$ feasible $w^* \neq 0$ infeasible

		0	0	0	0	1	1		
		x_1	x_2	x_3	x_4	x_5	x_6	b	Ratio
0	x_3	1	2	1	0	0	0	5	5/1 = 5
1	x_5	2	1	0	0	1	0	4	4/2 = 2
1	x_6	1	-1	0	1	0	1	1	1/2 = 0.5 ✓
	0	-3 ✓	0	0	-1	0	0		

$w = 1 \times 4 + 1 \times 1 = 5$

		0	0	0	0	1	1		
		x_1	x_2	x_3	x_4	x_5	x_6	b	Ratio
0	x_3	0	3	1	1	0	-1	4	4/3
1	x_5	0	3	0	2	1	-2	2	2/3 ✓
0	x_1	1	-1	0	-1	0	1	1	1/-1
	0	0	-3 ✓	0	-2	0	3		

$w = 1 \times 2 + 1 \times 0 = 2$

		0	0	0	0	1	1	
		x_1	x_2	x_3	x_4	x_5	x_6	b
0	x_3	0	0	1	-1	-1	1	2
0	x_2	0	1	0	2/3	1/3	-2/3	2/3
0	x_1	1	0	0	-1/3	1/3	1/3	5/3
	0	0	0	0	0	1	1	

$w = 0$ feasible

Phase II : min f continue table without x_5. x_6

		-1	-4	0	0	
		x_1	x_2	x_3	x_4	b
0	x_3	0	0	1	-1	2
-4	x_2	0	1	0	2/3	2/3
-1	x_1	1	0	0	-1/3	5/3
	0	0	0	0	7/3	

全正

$x^* = (\frac{5}{3}, \frac{2}{3}, 2, 0)$

$f^* = -1 \times \frac{5}{3} - 4 \times \frac{2}{3} = -\frac{13}{3}$